Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/28/2004
PI Name:	Faghri, Mohammad Ph.D.		
Project Title:	Phase Change in Low and Jittering Gravity Environment Simulated via Electromagnetic Field		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02881-0805	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	05/01/2000	**End Date:**	11/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note - end date changed from 11/01/2003		
Key Personnel Changes/Previous PI:	Faghri, M.		
COI Name (Institution):	Charmchi, M. (Univ. of Massachusetts-Lowell) Kalluri, D. K. M. (University of Massachusetts-Lowell)		
Grant/Contract No.:	NAG3-2426		
Performance Goal No.:			
Performance Goal Text:			
	Transport processes associated with melting of an electrically conducting Phase Change Material (PCM), placed inside a rectangular enclosure, under low-gravity environment, and in the presence of an electromagnetic field is simulated numerically and experimentally. Electromagnetic forces damp the natural convection as well as the flow induced by sedimentation and/or floatation, and thereby simulating the low gravity environment of outer space. Computational experiments are conducted for both sidewall heating and top-wall heating in both two- and three-dimensional rectangular enclosures. The governing equations are discretized using a control-volume-based finite difference scheme. Numerical solutions are obtained for true low-gravity environment as well as for the simulated-low-gravity conditions. The effects of electromagnetic field on the natural convection, solid phase floatation/sedimentation, liquid-solid interface location, solid melting rate, and flow patterns are investigated. It is found that the melting under low-gravity		

Task Description:	environment can reasonably be simulated on earth via applying a strong horizontal magnetic field. However, the flow patterns obtained for the true low-gravity cases are not similar to the corresponding cases solved for the simulated-low-gravity environment.On the other hand, it is shown that under electormagnetic field simulation, where the Lorentz force is caused by the transverse electric and magnetic fields, it is possible to minimize the flow field distortiion caused by the high magnetic field and therefore achieving a much better simulation of low-gravity. Both solidification and melting experiments were performed. These set of experiments are for sidewall phase change processes and therefore, addresses the fixed solid phase conditions. The effects of magnetic field on phase change rate and on the shape of the solid/melt interface are studied. The solid thickness is measured via ultrasonic techniques and the solid/melt interface is mapped using florescent light shadowgraphy through a transparent window. The collected data consist of temperature history, ultrasonic detection of the interface, florescent light shadowgraphy and solid phase volume fraction. The results showed that the presence of magnetic field had a marked effect on melting and natural convection whereas; phase change convection was noticeable in the solidification cases. The objective of the experimental part of the project is to design the Experimental Apparatus and Measurement System. Experiments will be conducted to observe solidification and melting processes under low gravity condditions induced by electromagnetic fields and to measure the temperature field and the solid thickness as a function of time and wall temperature conditions. The objective of the numerical part of the project is to model and solve the phase change problem under low gravity environment in two- and three-dimensional enclosure. Also, preliminary computations will be conducted to assist with the design of the experiment. Tasks Performed in the Numerical Part A (M. Faghri)Development of the computer codeValidating the code in the limiting casesFormulation of the ProblemSimulation of some cases to assist experiments. Also conduct computational experiments for both fixed(sidewall heating)and unfixed(top-wall heating)PCM in three-dimensional enclsoures. Tasks Performed in the Experiment Part B (M. Charmchi)Review of experiment design requirements: test cell dimensions, materials selections, boundary conditions, fabrication procedures, and assembly considerations.Instrumentation Requirements: data requirements and data acquisitions:Temperature measurements (sensors specifications and positions)Thermal environmental control (power requirements -heating and cooling- and equipment specifications)Data acquisition system (I/O boards, PC interface, and DAQ software). Conduct experiments to observe solidification and melting processes under low gravity conditions simulated by electromagnetic fields and measure temperature field and solid thickness as a funciton of time and the wall temperature conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	3D-Phase Change Simulated Low Gravity via Electromagnetic Fields (Sidewall): To date the work related to the sidewall heating problem is 100% complete. The manuscript has been accepted for presentation in the Summer 2004 Heat Transfer Conference sponsored by ASME and is under final review for publication with the International Journal of Numerical Methods in Heat and Fluid Flow. Contained within this manuscript are some new findings which are a result of the application of an electromagnetic field. Unique flow circulations are created with the application of an electromagnetic field. Furthermore, a flow reversal phenomenon was discovered which has significant influence on the solid / liquid interface shape and flow field. 3D-Phase Change Simulated Low Gravity via Electromagnetic Fields (Topwall): Since the last progress report (10/2003) the work has been focused primarily on the unfixed three-dimensional topwall heating problem. The problem formulation has been completed. Work related to the Fortran code has proved challenging in acquiring a converged solution with physically reasonably results. The addition to the third dimension to the code has proved to be more complicated than initially thought. A number of issues have been resolved and the code has been validated by 3D limiting cases which approximate previous 2D results. Currently cases are being run to establish the equivalent magnetic and electromagnetic field levels necessary to simulate low gravity. Once, these equivalent values have been determined the parametric studies will be completed. The remaining tasks to be completed on the topwall problem are as follows: 1. Establish equivalent field levels. 2. Complete parametric study. 3. Write draft of topwall problem. 4. Send paper out for publication. 3D-Phase Change Simulated Low and Jittering Gravity via Electromagnetic Fields (Sidewall): The initial stages of the 3D Sidewall Simulated Low and Jittering gravity problem have been completed. This consists of a thorough literature study to ensure the uniqueness on the problem. In addition the mathematical formulation has been completed for simulated jittering gravity via oscillating electric/magnetic fields.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Abstracts for Journals and Proceedings	Veilleux, D. L., Goncalves, E., Faghri, M., Asako, Y., Charmchi, M. "Phase change in a Three-Dimensional Rectangular Cavity Under Electromagnetically Simulated Low Gravity: Side Wall Heting" Proceedings of ASME HT-FED 2004, Charlotte, NC , , Jul-2004